(Mathematics-IV) Assignment - I (Unit-I)

Q.1 $(x^2-y^2-yz)p+(x^2-y^2-zx)q = z(x-y)$

Q.2 $(p^2+q^2)y = qz$ Solve by Charpit's Method.

Q.3 Solve the Linear Partial differential Equations.

$$\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial^2 z}{\partial x \partial y} = \sin x \cos 2y$$

Q.4 Solve the following PDE

$$(D^3-4D^2D'+5DD'^2-2D'^3)z = e^{y+2x}+(y+x)^{1/2}$$

Q.5 Solve $(r-6s+9t)z = 12x^2+36xy$

Q.6 Solve $(2D^2+5DD'+3D'^2)z = ye^x$

Q.7 Solve $(r-4s+4t+p-2q = e^{x+y}\cos(2x+3y)$

Q.8 Solve $D(D+D'-1)(D+3D'-2)z = x^2-4xy+2y^2$

Q.9 Solve $x^2\frac{\partial^2 z}{\partial x^2} - y^2\frac{\partial^2 z}{\partial y^2} = xy$

Q.10 $u_x+u_y = \sinh x+\sinh y$; $u(0,y) = 1+\cosh y$

Q.11 Find $(D^2 + DD' - 6D'^2)z = \cos(2x+y)$

Q.12 obtain a Partial Differential Equation that governs the family of surfaces.

$$z = (x-\alpha)^2 + (y-\beta)^2$$

Q.13 Solve By Lagrange's subsidiary / Auxilary Equation

$$(y^2 + z^2 - x^2)p - 2xyq + 2xz = 0$$

Q.14 Solve $r - t + p - q = 0$

Q.15 Solve $4r - 4s + t = 16 \log(x+2y)$